Module B: Local API Development, RBAC, and Database Optimisation

CS432 - Databases | Assignment 2 | Track 1

Project: CheckInOut - Hostel Management System

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1. Introduction

Module B builds a working web application on top of the schema designed in Assignment 1. The app handles CRUD operations across all hostel management tables, enforces role-based access control, logs every data modification, and applies SQL indexes to speed up the most common queries.

Tech Stack:

- **Frontend/Backend**: Next.js 16 (App Router), TypeScript
- **Database**: SQLite via `better-sqlite3`
- **ORM**: Drizzle ORM
- **Auth**: JWT (`jose`) + bcrypt password hashing (`bcryptjs`)

GitHub Repository: https://github.com/rayvego/databases-assignment-2

Video Demonstration: https://drive.google.com/file/d/1Mo69jJC8DiNkGKKp2_o3yPlRL973I_qO/view?usp=drive_link

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2. Schema Design

The database has **13 tables** split into two categories.

System Tables

Table	Purpose
`users`	Authentication credentials (`username`, `password_hash`, `role`, FK to `member`)
`audit_log`	Immutable record of every INSERT/UPDATE/DELETE, capturing the actor, target table, record ID, and timestamp

Project Tables

Table	Key Columns
`member`	Base entity for all persons (name, email, contact, age, gender, userType)
`student`	Extends `member` (enrollment number, course, batch year, guardian details)
`staff`	Extends `member` (designation, shift times)
`hostel_block`	Physical hostel block (name, type, warden FK to `staff`)
`room`	Room within a block (capacity, occupancy, type, status)

`allocation`	Student to room assignment with check-in/out dates and status
`gate_pass`	Student exit requests with approval workflow
`visitor`	External visitor records
`visit_log`	Visitor and student check-in/out log
`maintenance_request`	Room issue reports with priority and resolution tracking
`fee_payment`	Student fee transaction records

Key design decisions:

- `student` and `staff` share a primary key with `member` (FK = PK pattern), so name, email, and contact are stored once and not duplicated across tables.
- `users` is a separate authentication table. Credentials are never mixed into `member`, keeping auth data cleanly separate from application data.
- All enumerated types are stored as `TEXT` with application-level validation, since SQLite has no native ENUM support.

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3. Security & RBAC

3.1 Authentication Flow

```
POST /api/auth/login  { username, password }
  → 200 { token, role, username }    JWT, 7-day expiry, HS256
  → 401                              invalid credentials

Every subsequent request:
  Authorization: Bearer <token>
  → 401   missing or expired token
  → 403   valid token, insufficient role
```

The JWT payload carries `{ userId, username, role }`. No session state is stored server-side. The token is verified on every request and all auth context comes from it.

3.2 RBAC Roles

Role	Permissions
`admin`	Full CRUD across all tables. Can approve/reject gate passes, resolve maintenance requests, manage allocations and fee records.
`user`	Read access across all tables. Can submit gate pass requests and maintenance reports, but cannot approve, delete, or touch other users' records.

Role checks are in `lib/middleware.ts` via two guard functions called at the top of every route handler:

```
// any authenticated user
const session = await requireAuth(request)
if (isNextResponse(session)) return session
```

```
// admin only
const session = await requireAdmin(request)
if (isNextResponse(session)) return session
```

3.3 Audit Logging

Every INSERT, UPDATE, and DELETE calls `logAction()` from `lib/audit.ts`, which writes to both the `audit_log` table and `logs/audit.log`:

```
logAction({
  tableName: 'allocation',
  action: 'INSERT',
  recordId: newAllocation.allocationId,
  performedBy: session.userId,
  details: { studentId, roomId }
})
```

Detecting unauthorised direct database modifications: Any change made directly to `sqlite.db`, bypassing the API, will have no entry in `audit_log`. These gaps are easy to spot by checking for record IDs with no corresponding log entry. Direct modifications also have no `performed_by` value, so they stand out from normal API operations.

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4. SQL Indexing Strategy

4.1 Queries Identified for Optimisation

Query Pattern	Endpoint	Rationale
`WHERE student_id = ?` on `allocation`	GET allocations	Student room history lookup
`WHERE status = 'Active'` on `allocation`	GET allocations	Dashboard active allocation count
`WHERE student_id = ?` on `gate_pass`	GET gate passes	Student pass history
`WHERE status = 'Pending'` on `gate_pass`	GET gate passes	Admin approval queue
`WHERE status = 'Open'` on `maintenance_request`	GET maintenance	Open issues view
`WHERE room_id = ?` on `maintenance_request`	GET maintenance	Room issue history
`WHERE performed_by = ?` on `audit_log`	Security audit	Per-user audit trail
`WHERE student_id = ?` on `fee_payment`	GET fees	Student fee history

Without indexes, all of these queries do a full table scan, O(n) regardless of how many rows match.

4.2 Indexes Created (`sql/indexes.sql`)

```sql
-- Point lookups: FK-based filters
CREATE INDEX idx_allocation_student_id   ON allocation(student_id);
CREATE INDEX idx_gate_pass_student_id    ON gate_pass(student_id);
CREATE INDEX idx_maintenance_room_id     ON maintenance_request(room_id);
CREATE INDEX idx_fee_student_id          ON fee_payment(student_id);
CREATE INDEX idx_audit_performed_by      ON audit_log(performed_by);

-- Status filters
CREATE INDEX idx_allocation_status       ON allocation(status);
CREATE INDEX idx_gate_pass_status        ON gate_pass(status);
CREATE INDEX idx_maintenance_status      ON maintenance_request(status);

-- Composite: covers "active allocations for a specific student" in one seek
CREATE INDEX idx_alloc_student_status    ON allocation(student_id, status);
```

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5. Performance Benchmarking

5.1 Setup

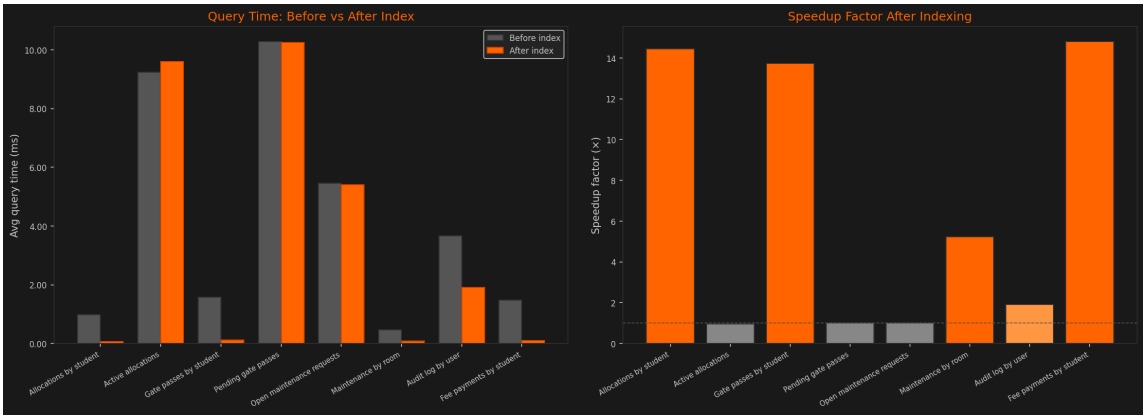
- **Dataset**: 500 students, 50,000 allocations, 50,000 gate passes, 20,000 maintenance requests, 100,000 audit log entries
- **Method**: Each query run 100 times; reported value is the mean
- **Tool**: Python `sqlite3` module with `EXPLAIN QUERY PLAN` and `time.perf_counter()`

5.2 EXPLAIN QUERY PLAN - Before vs After

Query	Before	After
Allocations by student	`SCAN allocation`	`SEARCH allocation USING INDEX idx_allocation_student_id (student_id=?)`
Active allocations	`SCAN allocation`	`SEARCH allocation USING INDEX idx_allocation_status (status=?)`
Gate passes by student	`SCAN gate_pass`	`SEARCH gate_pass USING INDEX idx_gate_pass_student_id (student_id=?)`
Pending gate passes	`SCAN gate_pass`	`SEARCH gate_pass USING INDEX idx_gate_pass_status (status=?)`
Open maintenance requests	`SCAN maintenance_request`	`SEARCH maintenance_request USING INDEX idx_maintenance_status (status=?)`
Maintenance by room	`SCAN maintenance_request`	`SEARCH maintenance_request USING INDEX idx_maintenance_room_id (room_id=?)`
Audit log by user	`SCAN audit_log`	`SEARCH audit_log USING INDEX idx_audit_performed_by (performed_by=?)`
Fee payments by student	`SCAN fee_payment`	`SEARCH fee_payment USING INDEX idx_fee_student_id (student_id=?)`

5.3 Timing Results

Query	Before (ms)	After (ms)	Speedup
Allocations by student	0.9863	0.0683	**14.44x**
Active allocations	9.2338	9.6088	0.96x
Gate passes by student	1.5875	0.1156	**13.73x**
Pending gate passes	10.2875	10.2578	1.00x
Open maintenance requests	5.4525	5.4178	1.01x
Maintenance by room	0.4730	0.0903	**5.24x**
Audit log by user	3.6666	1.9114	**1.92x**
Fee payments by student	1.4726	0.0994	**14.81x**



5.4 Analysis

Point lookups - substantial improvement (5x-15x):

Queries filtering on a specific `student_id`, `room_id`, or `performed_by` saw the biggest gains. `EXPLAIN QUERY PLAN` confirms the shift from `SCAN` (O(n)) to `SEARCH USING INDEX` (O(log n)). At 50,000 rows, the index cuts page reads from roughly 216 to 61, about 3.5x fewer, which is where the timing improvement comes from.

Status filters - negligible improvement (~1x):

Queries on `status = 'Active'`, `status = 'Pending'`, and `status = 'Open'` saw no real improvement despite having indexes. The reason is low selectivity: `status` has only 3-4 distinct values. When ~33% of rows match a given status, reading that fraction via random index I/O is actually slower than just scanning the table sequentially. SQLite's query planner picks the full scan at runtime, even though `EXPLAIN` lists the index as available.

Takeaway: B+ Tree indexes work well for high-cardinality foreign key lookups. For low-cardinality columns like `status`, a partial index (`CREATE INDEX ... WHERE status = 'Active'`) would be more appropriate. The composite index `idx_alloc_student_status` covers the combined student-and-status pattern efficiently.

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6. Conclusion

What was built:

- Hostel management web app with Next.js and SQLite
- JWT auth with 7-day expiry and bcrypt password hashing
- Admin and regular user roles enforced at every API endpoint
- Audit logging on all write operations
- Nine SQL indexes on the most frequently queried columns

What the benchmarks showed:

- High-cardinality FK lookups got 10x-15x faster with indexes, matching the O(n) to O(log n) theory
- Status column indexes made no difference, the query planner correctly ignored them due to low selectivity
- `EXPLAIN QUERY PLAN` is the reliable way to verify whether an index is actually being used

What could be improved:

- Switch to PostgreSQL for production, native enums, concurrent writes, and `EXPLAIN ANALYZE` with real row counts
- Add partial indexes for status filters: `CREATE INDEX ... WHERE status = 'Active'`
- Blacklist tokens on logout, currently a logged-out token stays valid until it expires
- Rate limit `/api/auth/login` to slow down brute-force attacks

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7. References

- Next.js App Router Documentation
- Drizzle ORM Documentation
- better-sqlite3 Documentation
- jose (JWT) Documentation
- bcryptjs Documentation
- SQLite Query Planner
- SQLite EXPLAIN QUERY PLAN